

1 Best case and worst case

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Linear Search

$$A = [6 | 12 | 9 | 7 | 4 | 3 | 16]$$

key = 7 = 5
key not found in array
so we go through array

Best case $\Rightarrow$ if we get 1st element

$$B(n) = O(1)$$

Worst case = when key take max time
and key found in last index.

$$W(n) = O(n)$$

$$\text{Average case} = \frac{\text{all possible case time}}{\text{no. of cases}}$$

its not always possible

index 1 = 1
index 2 = 2
index 3 = 3 comparison

$$\begin{aligned} \text{Avg time} &= \frac{1 + 2 + 3 + \dots + n}{n} \\ &= \frac{n(n+1)}{2n} = \frac{n+1}{2} \end{aligned}$$

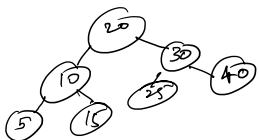
$$T(n) = \frac{n+1}{2}$$

Asymptotic Notation

$$\begin{aligned} B(n) &= 1 & W(n) &= n \\ B(n) &= O(1) & W(n) &= O(n) \\ B(n) &= \Omega(1) & W(n) &= \Omega(n) \\ B(n) &= \Theta(1) & W(n) &= \Theta(n) \end{aligned}$$

Notations can be written for any case
notation and cases are not related
notations are for functions

Binary Search Tree



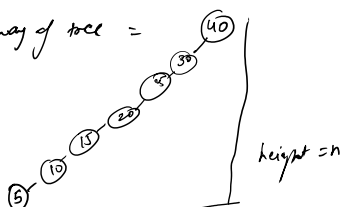
to find 15
we have done 3 comparison
which is height of tree balanced
height of tree = $\log n$

Best case = Root element
Best case time = $B(n) = 1$

Worst case = Searching for leaf element

Worst case time = height of tree

Another way of tree =



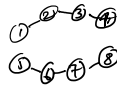
$$\begin{aligned} \text{Worst case} &= \min W(n) = \log n \\ &= \max W(n) = n \end{aligned}$$

min and max in worst case depend on type
of data is structured

Disjoint Sets

- used to detect cycle in graph
- Kruskal algo used to detect cycle in graph
- Disjoint sets are sets which don't have anything common

e.g. $S_1 = \{1, 2, 3, 4\}$
 $S_2 = \{5, 6, 7, 8\}$
 $S_1 \cap S_2 = \emptyset$

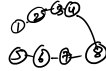


operations → find
union

- find means find the element in which set it belongs to
- union

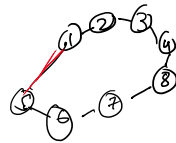
$S_1 = \{1, 2, 3, 4\}$
 $S_2 = \{5, 6, 7, 8\}$

(4, 8)

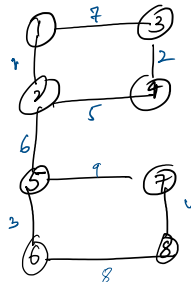


$S_1 \cup S_2 = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$S_3 = \{1, 2, 3, 4, 5, 6, 7, 8\}$
 (1, 5)



if elements are present in same set then there is cycle



$U = \{1, 2, 3, 4, 5, 6, 7, 8\} = \text{universal set}$

(1, 2) lets form a set from universal set

$S_1 = \{1, 2\}$

$S_2 = \{3, 4\}$

$(5, 6) = S_3 = \{5, 6\}$

$(7, 8) = S_4 = \{7, 8\}$

find (2, 4) they are in diff set so perform union

$S_5 = \{1, 2, 3, 4\}$ now S_1 and S_2 are destroyed

find (2, 5)

union S_5 and S_3

$S_6 = \{1, 2, 3, 4, 5, 6\}$

= now S_3 and S_5 are destroyed

find (1, 3)

both belongs to S_6
 that means cycle is there
 now lets not include

find (6, 8)

both belongs to different set

$S_7 = \{1, 2, 3, 4, 5, 6, 7, 8\}$

= S_6 and S_4 are destroyed
 now we have only one set S_7

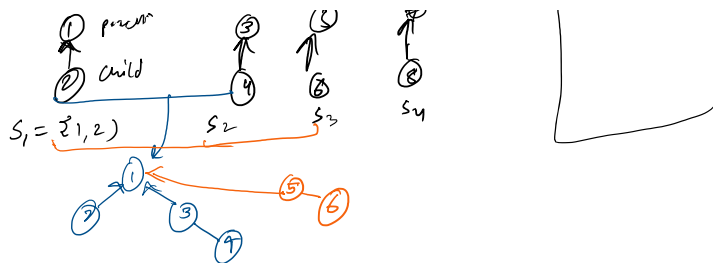
find (5, 7)

both belong to same set
 means cycle is there

lets show sets graphically.

$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

diagram



How can we do using Array

Parent

-1	-1	-1	-1	-1	-1	-1	-1
1	2	3	4	5	6	7	8

= (1, 2)
both are from diff set so union

-2	-1	-1	-1	-1	-1	-1	-1
1	2	3	4	5	6	7	8

= (3, 4)

-2	-1	-2	-1	-1	-1	-1	-1
1	2	3	4	5	6	7	8

-2 means that 2 nodes are there

= 5, 6

= 7, 8

-2	-1	-2	-1	-2	-1	-2	-1
1	2	3	4	5	6	7	8

= (2, 4)

both are diff parent perform Union

-4	-1	-1	-3	-1	-1	-1	-1
1	2	3	4	5	6	7	8

= 2, 5

-6	-1	-1	-3	-1	-5	-1	-1
1	2	3	4	5	6	7	8

collapsing find is when we find that
element is having parent then put it there

ex



now we find parent of 6 is 1
then we can direct put under 1

